



Chan Lee

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🔗 GitHub Profile

🌐 Tech Blog

EDUCATION

- **Korea University** Mar. 2022 – Feb. 2028 (Expected)
College of Informatics / Computer Science and Engineering GPA: 3.90 / 4.5 (94.0 / 100)

PROJECTS

- **Lucid — Deep Learning Framework from Scratch** *Python, NumPy, Autodiff* 🔗 [Source Code](#)
 - Built a mini-PyTorch-style deep learning framework entirely from scratch in Python, featuring a dynamic **computational graph** and reverse-mode **automatic differentiation (backpropagation)**.
 - Developed a deep, hands-on understanding of how computational graphs, the chain rule, and gradient propagation work under the hood; released the package on PyPI as `lucid-dl`.
- **Hybrid Hard-Negative Mitigation for Dense Retrieval** *NLP Term Project*
 - Co-authored a research report for the Natural Language Processing course (COSE461), tackling hard-negative confusion in dense retrieval on a frozen ColBERT-v2 backbone across three BEIR benchmarks.
 - Owned the **Lightweight Model Intervention**: designed and implemented **anchored LoRA** — a low-rank adapter (~0.3% of parameters) across all layers with a per-token representation anchor that protects already-correct query embeddings.
 - Achieved a strict **+0.030 NDCG@10** improvement on SciFact over the frozen baseline (3/3 seeds), retaining 88% of plain LoRA's hard-query gain while cutting its easy-query loss by 75%.
- **Attention U-Net for Medical Image Segmentation** *Deep Learning, Computer Vision* 🔗 [Source Code](#)
 - Reproduced Attention U-Net (Oktay et al., 2018) for medical image segmentation and trained it on the Kvasir-SEG dataset.
 - Ran the entire model on my own Lucid framework, validating its autograd and neural-network layers end-to-end on a real computer-vision task.
- **NCSN — Score-Based Generative Modeling** *PyTorch, Generative Models* 🔗 [Source Code](#)
 - Implemented a Noise Conditional Score Network (NCSN) for score-based generative modeling in PyTorch.
 - Studied and implemented score matching with annealed Langevin dynamics sampling across multiple noise scales.

TECHNICAL SKILLS AND INTERESTS

Languages: Python (primary), C/C++

ML / DL Libraries: PyTorch, NumPy, MLX, Pandas, Matplotlib

Major Coursework: Machine Learning, Reinforcement Learning, Artificial Intelligence, Natural Language Processing, Data Science, Probability & Random Processes, Data Structure, Algorithms, Operating Systems, Computer Architecture, Computer Network, Theory of Computation, Discrete Mathematics, Mathematics for Computer Science I & II, Computer Programming I & II

Areas of Interest: Computer Vision, Generative Models (Diffusion/Flow Models), World Models

Currently Studying: Independently studying the theory of diffusion models from Lai et al., *The Principles of Diffusion Models* (arXiv:2510.21890, 2025).

ACHIEVEMENTS & ACTIVITIES

- **President**, AIKU (Korea University Deep Learning Student Academic Society) — 8th Cohort
- **3rd Place**, Korea Univ. × Yonsei Univ. × Upstage Hackathon — preventing sensitive-information (PII) exposure in LLM RAG systems
- **Technical Blog**, Velog (@lumerico284) — writing about self-directed deep learning study